

by automating anomaly detection, which will cut waste and avoid budget overruns. Predictive analytics will also be made easier by AI-powered solutions, which will assist teams in proactively allocating resources and planning capacity.

Additionally, it will be easier to integrate cloud cost management tools with larger FinOps frameworks. A culture of continuous cost optimisation will be fostered as companies embrace comprehensive FinOps strategies, with an emphasis on coordinating engineering practices with financial accountability. A more comprehensive view of cloud spending and efficient policy enforcement will be made possible by the anticipated tighter integrations with budgeting, procurement, and governance platforms in future tools.

Effective cloud cost management has many advantages, chief among them being control, predictability,

and transparency. Organisations can clearly see how much they are spending on the cloud across all projects and departments when there is transparency. Stakeholders can comprehend precisely where resources are being used and how expenses mount up with the help of comprehensive cost breakdowns and real-time insights. This clarity fosters accountability, enables more informed decision-making, and helps identify wastage or inefficiencies promptly. Improved communication between technical teams and financial stakeholders is another benefit of transparent cost data.

Another important advantage is predictability. Organisations can more

accurately predict future costs by utilising forecasting tools, anomaly detection, and historical data. Better resource planning and budgeting are made possible by this foresight, which also lowers the financial risks and surprises brought on by erratic cloud bills. The ability to establish budgets, implement rules, and maximise resource use gives users control over their cloud spending.

In conclusion, organisations can better manage cloud costs by combining transparency, predictability, and control. This ensures financial accountability, operational efficiency, and the capacity to scale with confidence while staying within budgetary constraints. **END** 

By: Sufyan bin Uzayr

The author, apart from being a writer, is an educator and entrepreneur. He holds multiple degrees in management, information technology and humanities. He has written several books and is currently the director of a multinational IT company. As an educator, he has taught at universities and institutions worldwide.

■ ■ Continued from page....53

Technical contributions include Pi-specific tuning (GPU memory set to 128MB, CPU governor set to performance), encapsulating detection and alert logic in dual-container pods, and integrating a unified observability stack (Prometheus, Loki, Grafana) for real-time metrics and logs. In our experiments, the automated workflow was 40% faster than manual setup while maintaining low node resource usage. We identified performance bottlenecks (fixed-interval frame polling, static resource limits) and outlined future work on adaptive frame-rate handling, event-driven alerts, and Kubernetes vertical pod autoscaling for dynamic resource tuning. Overall, this work serves as a blueprint for bringing robust CI/CD and observability best practices to latency-sensitive edge deployments, facilitating the transfer of cloud-native methodologies to resource-constrained environments.

While evaluated for violence detection, our framework solves three fundamental edge challenges: ARM resource partitioning, bandwidth-constrained observability, and resilient CI/CD for unstable

networks. The K3s modifications and design patterns can be extended to any latency-sensitive edge deployment, offering a proven blueprint for combining cloud-native tooling with edge constraints. **END** 

References

- R. Labs, 'K3s: Lightweight Kubernetes for IoT and Edge', <https://k3s.io/>, 2020
- A. Zhang and etc, 'Benchmarking K3s for Kubernetes at the Edge', IEEE Access, 2021
- B. Sharma and M. Singh, 'Automating Edge Device Provisioning with Ansible', in Proc. ACM SIGBED, 2020
- C. Lee and D. Kim, 'Integrating Jenkins-based CI/CD with Docker Swarm', Journal of Systems and Software, 2022
- B. Soundararajan, 'Monitoring with Prometheus: A Case Study', 2018, Prometheus documentation and whitepaper
- S. Thakur and R. Das, 'Prometheus and Grafana for IoT Monitoring', IJCSN, 2021
- G. Labs, 'Loki: How it works', <https://grafana.com/docs/loki/latest/>, 2020
- M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, 'Mobilenetv2: Inverted residuals and linear bottlenecks', in Proc. CVPR, 2018
- S. Hochreiter and J. Schmidhuber, 'Long short-term memory', Neural Computation, 1997
- GitHub repository for reference: https://github.com/harshchandrakar/violence_detection_infra

By: Harsh Kumar Chandrakar and Thangaraju B.

The authors are associated with the Open Source Technology Lab at the International Institute of Information Technology, Bengaluru (IIITB).